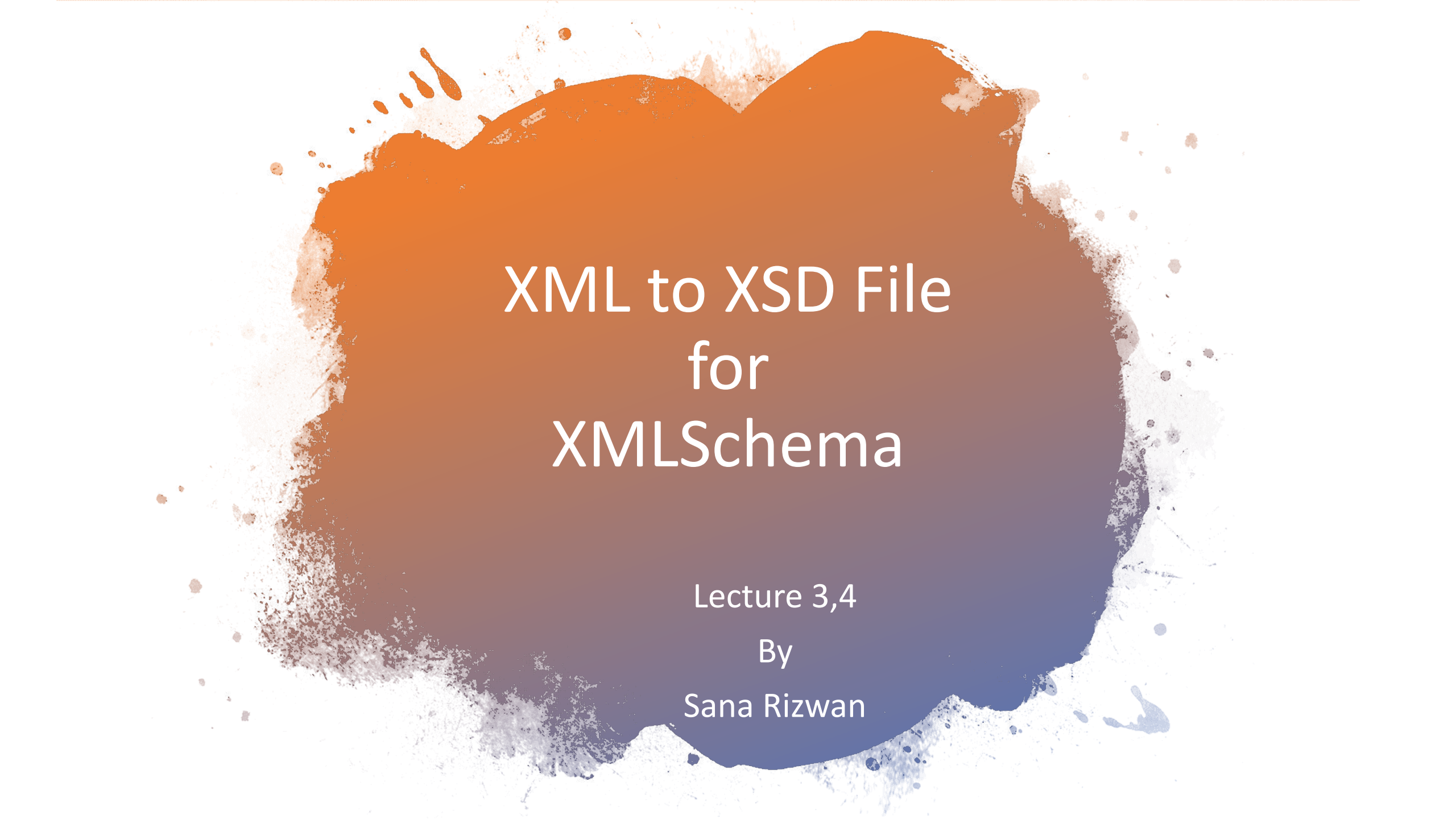


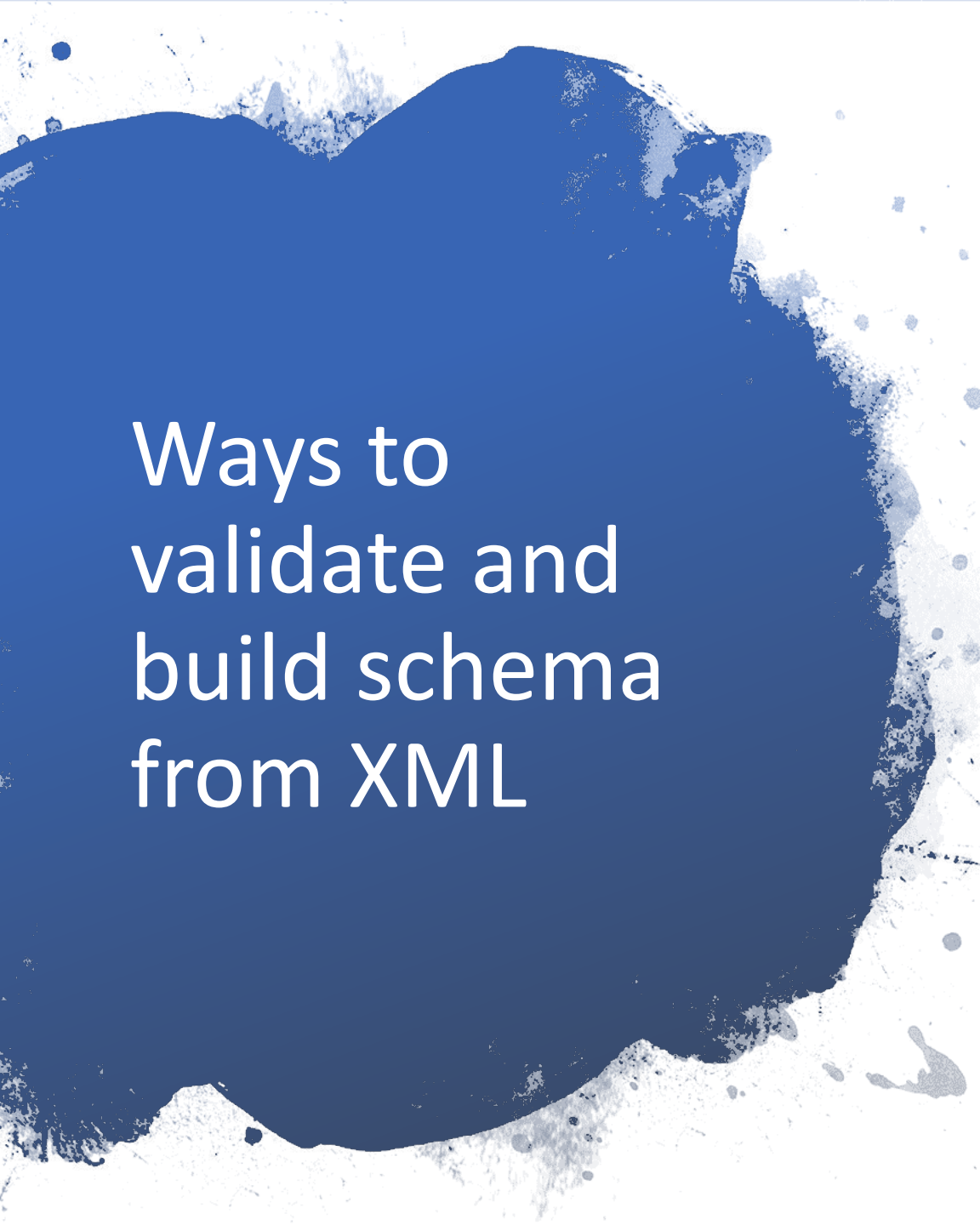


XML to XSD File for XMLSchema

Lecture 3,4

By

Sana Rizwan



Ways to validate and build schema from XML

- Well Form
- DTD
- XSD—XML Schema Definition

How to: Create an XML schema from an XML document

- The XML editor lets you create an XML Schema definition language (XSD) schema from an XML document. The XML file determines how the schema is generated in the following manner:
- If the XML document has no schema or Document Type Definition (DTD) associated with it, the data in the XML document is used to infer a new XML Schema.
- If the XML document contains an associated DTD, the external DTD and internal subset are converted to a corresponding XML Schema.
- If the XML document contains an inline XML-Data Reduced (XDR) schema, the XDR schema is converted to a corresponding XML Schema.

XSD Building Block

- Elements and attributes that may appear in the document.
- Number and order of children.
- Data types for elements and attributes.
- Defaults and fixed values for elements and attributes.

Why XSD?

- XML based
 - Use your XML editor to make an XML schema.
- More common than DTD
- Can specify type.
 - Allowable content
 - Validate correctness.
 - Define facets, or restrictions on data.
 - Define patterns (formats)
 - Convert between different types.
- Easy to communicate across parties, because expectations are properly set.
 - How to format date, time, currency, etc.

XML Root Element

- To reference a schema, the root element requires several items:
 - *xmlns default namespace*
 - *xmlns:xsi*
http://www.w3.org/2001/XMLSchema-instance
 - *xsi:schemaLocation Where your XSD file is.*

```
C:\Users\ucinf\Documents\uc\17FSIS7024\Assignment\users_xmlschema.xml - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window ?
1 <?xml version="1.0"?>
2 <?xml-stylesheet type="text/xsl" href="users.xsl"?>
3 <userdetails xmlns:f="http://www.w3.org/2001/XMLSchema" xmlns:xsi="
  http://www.w3.org/2001/XMLSchema-instance"
4   xsi:noNamespaceSchemaLocation="uservalidation.xsd">
5   <users>
6     <firstname>Brandan</firstname>
7     <lastname>Jones</lastname>
8     <bearcatid>jonesbr</bearcatid>
9     <favoriteshortcut>Control Shift T in Chrome</favoriteshortcut>
10  </users>
11  <users>
12    <firstname>Checkers</firstname>
13    <lastname>Jones</lastname>
14    <bearcatid>jonescj</bearcatid>
15    <favoriteshortcut>Control W in Chrome</favoriteshortcut>
16  </users>
17 </userdetails>
```

XML Document

XSD Document

```
C:\Users\ucinf\Documents\uc\17FSIS7024\Assignment\uservalidation.xsd - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window ?
1 <?xml version="1.0" encoding="UTF-8"?>
2 <xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:element name="userdetails">
4     <xs:complexType>
5       <xs:sequence>
6         <xs:element name="users" maxOccurs="unbounded">
7           <xs:complexType>
8             <xs:sequence>
9               <xs:element name="firstname" type="xs:string"/>
10              <xs:element name="lastname" type="xs:string"/>
11              <xs:element name="bearcatid" type="xs:string"/>
12              <xs:element name="favoriteshortcut" type="xs:string"/>
13            </xs:sequence>
14          </xs:complexType>
15        </xs:element>
16      </xs:sequence>
17    </xs:complexType>
18  </xs:element>
19 </xs:schema>
```

Simple vs. Complex Types

- Complex types - `complexType`
 - Allow elements in their content
 - May carry attributes
- Simple types - `simpleType`
 - Cannot have element content
 - Cannot carry attributes

Simple Type

- `<car>BMW</car>`
- `<truck color> blue</truck color>`
- `<xsd:element name="car" type="string"/>`

Complex Type

<parent>

<father name="ali" profession="teacher" department="Maths">

</father>

<!--Simple Type-->

<son> musa</son>

</parent>

```

1  <?xml version="1.0"?>
2  <?xml-stylesheet type="text/xsl" href="users.xsl"?>
3  <userdetails xmlns:f="http://www.w3.org/2001/XMLSchema"
4  http://www.w3.org/2001/XMLSchema-instance"
5  xsi:noNamespaceSchemaLocation="uservalidation.xsd">
6  <users>
7    <firstname>Brandan</firstname>
8    <lastname>Jones</lastname>
9    <bearcatid>jonesbr</bearcatid>
10   <favoriteshortcut>Control Shift T in Chrome</favo
11 </users> I
12 <users>
13   <firstname>Checkers</firstname>
14   <lastname>Jones</lastname>
15   <bearcatid>jonescj</bearcatid>
16   <favoriteshortcut>Control W in Chrome</favoritesh
17 </users>
</userdetails>

```

Simple Types Available

Simple Element

- Contains only text. No other attributes or elements.
- Text: can be boolean, string, date, etc., or a custom type.
- Can add restrictions to limit data content.
 - Example: `<xs:element name="foo" type="xs:string"/>`

– Types available:

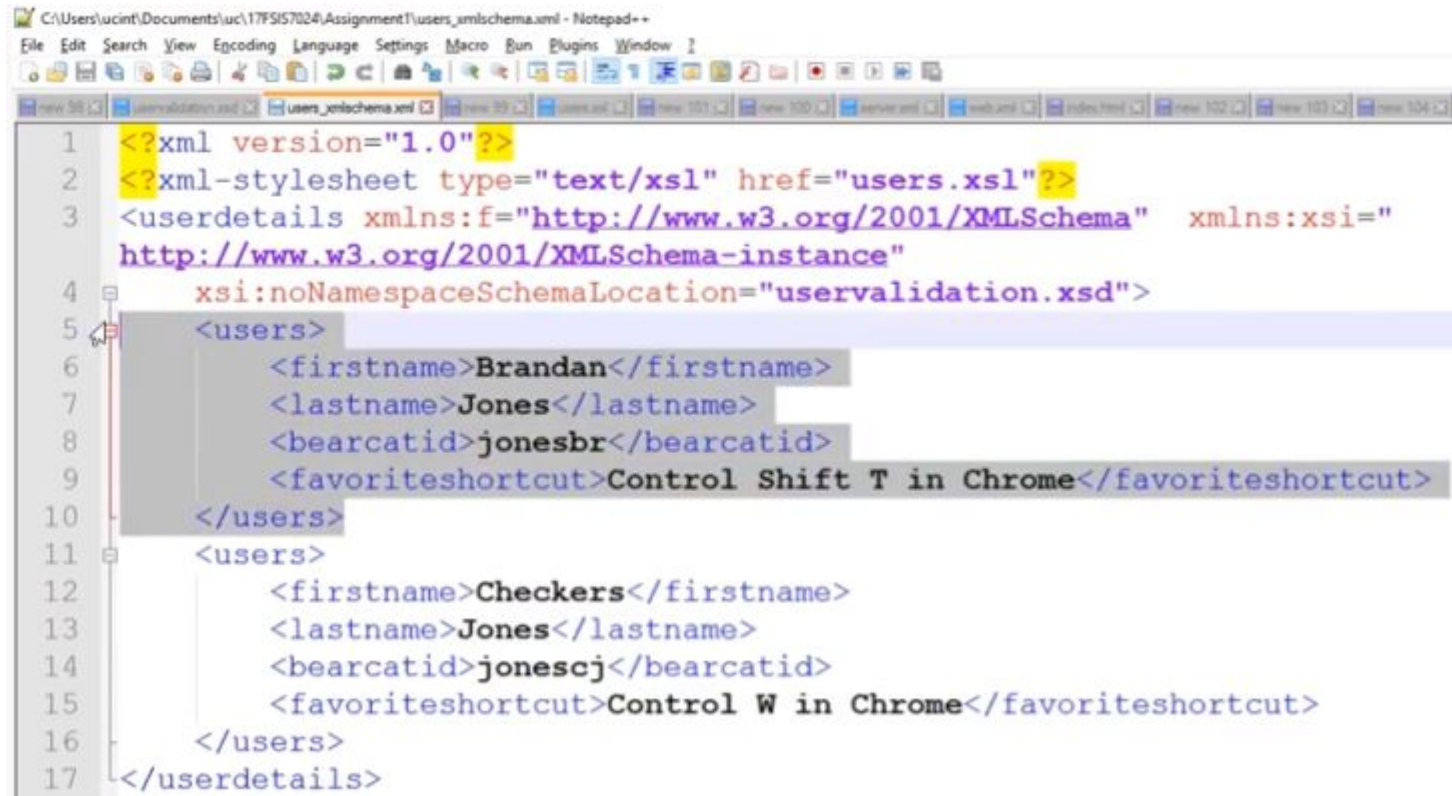
- xs:string
- xs:decimal
- xs:integer
- xs:boolean
- xs:date
- xs:time

– Other attributes:

- default: assigned to element when no other value is specified.
- fixed: automatically assigned to element; you cannot choose another.

Simple Types – Only have some text in the tags, can't have elements. It can have attributes but can't have sub-elements.

Complex Types– Can have elements in their contents and may have attributes.



```
1 <?xml version="1.0"?>
2 <?xml-stylesheet type="text/xsl" href="users.xsl"?>
3 <userdetails xmlns:f="http://www.w3.org/2001/XMLSchema" xmlns:xsi="
4 http://www.w3.org/2001/XMLSchema-instance"
5 xsi:noNamespaceSchemaLocation="uservalidation.xsd">
6 <users>
7 <firstname>Brandan</firstname>
8 <lastname>Jones</lastname>
9 <bearcatid>jonesbr</bearcatid>
10 <favoriteshortcut>Control Shift T in Chrome</favoriteshortcut>
11 </users>
12 <users>
13 <firstname>Checkers</firstname>
14 <lastname>Jones</lastname>
15 <bearcatid>jonescj</bearcatid>
16 <favoriteshortcut>Control W in Chrome</favoriteshortcut>
17 </users>
</userdetails>
```

Simple Types

- String
 - Token
 - Byte
 - unsignedByte
 - Integer
 - positiveInteger
 - Int (larger than integer)
 - unsignedInt
 - Long
 - Short
 - ...
- Time
 - dateTime
 - Duration
 - Date
 - ID
 - IDREF
 - IDREFS

Facets of Simple Types

- Facets = additional properties restricting a simple type
- 15 facets defined by XML Schema

Examples

- length
- minLength
- maxLength
- pattern
- enumeration
- whiteSpace
- maxInclusive
- maxExclusive
- minInclusive
- minExclusive
- totalDigits
- fractionDigits

Facets of Simple Types

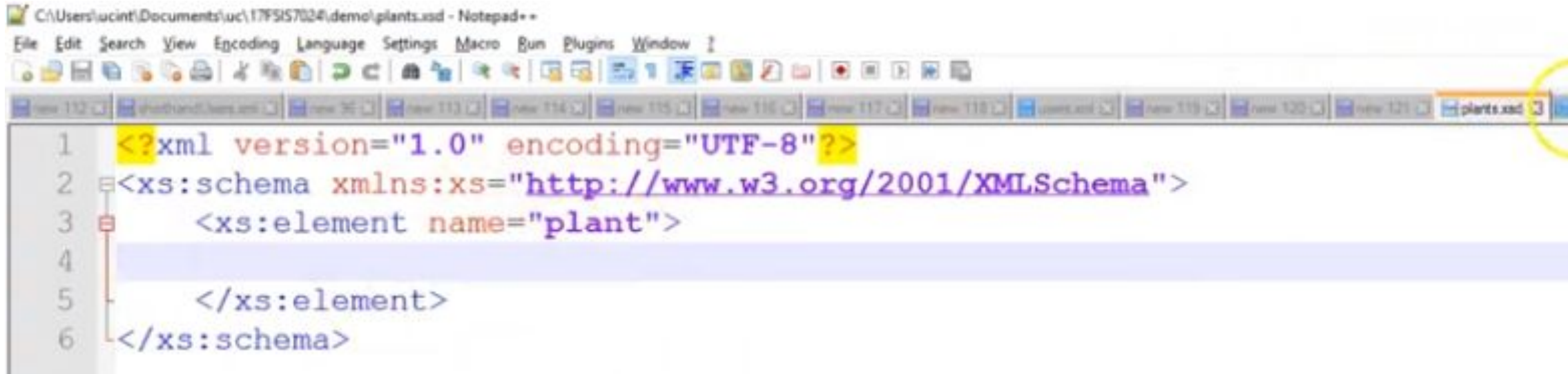
- Can further restrict a simple type by changing some facets
- Restriction = subset

XML Definition Document -- PlantPlaces

PlantPlaces.com accepts plant data in the following forms.

Element/Attribute Name	Description	Required?	Default Value	Type/Restrictions
plant	Root element	1		
_genus	Genus	1..*		Max Length 60
_species	Species	no	spp.	Max Length 60
_cultivar	Cultivated variety (Fuji apple, Golden Delicious apple, etc.)	no		Max Length 60
_height	height, in meters	no	0	integer, max: 100
_begin_bloom_date	date at which the plant begins blooming.	no		
_end_bloom_date	date at which blooms stop	no		
_edible	Does the plant contain edible parts?	no		Boolean
_sun_tolerance	How does the plant tolerate sun?	no		Enumeration: full_sun, part_sun, part_shade, full_shade
_bloom_color	color of flower	no		Enumeration: red, orange, yellow, blue, violet, white
_specimens	Contains a repeating group of specimens.	no		
__specimen	represents an individual specimen	1..*		

Create .xsd file



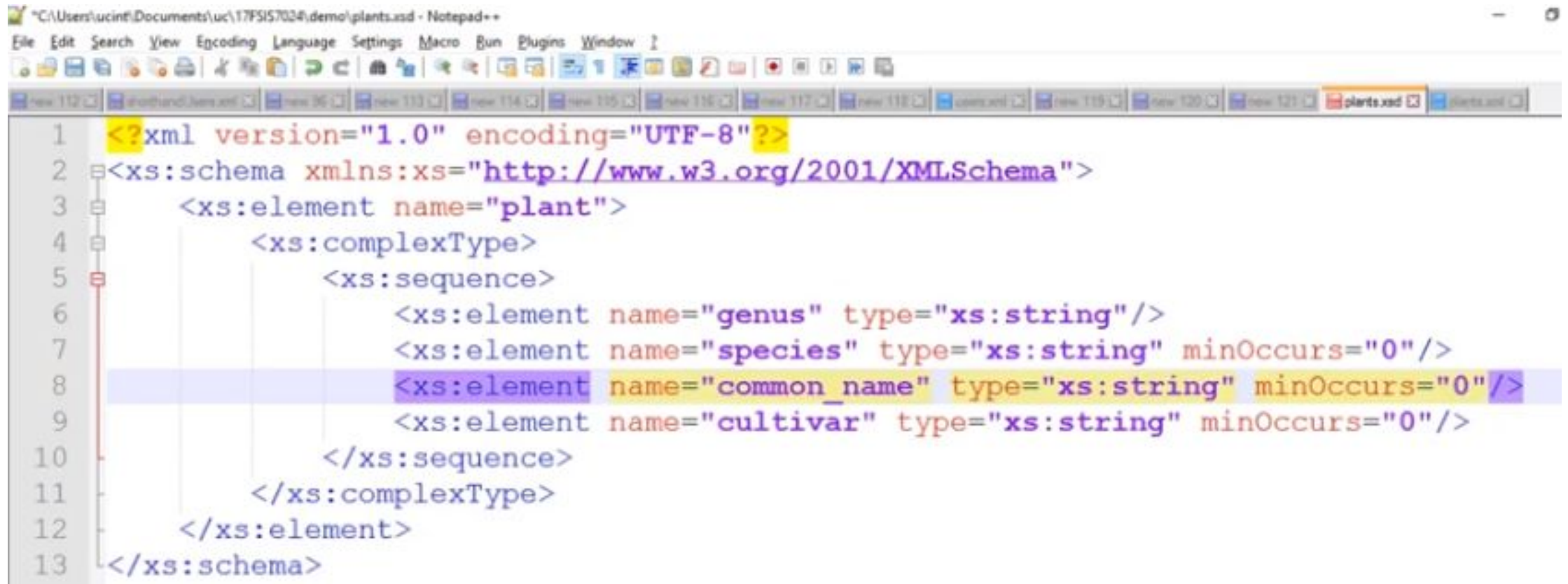
```
C:\Users\lucint\Documents\uc\17FSIS7024\demo\plants.xsd - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window
new 112 plants.xsd new 113 new 114 new 115 new 116 new 117 new 118 users.xml new 119 new 120 new 121 plants.xsd
1 <?xml version="1.0" encoding="UTF-8"?>
2 <xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:element name="plant">
4
5   </xs:element>
6 </xs:schema>
```

Create .Xml file



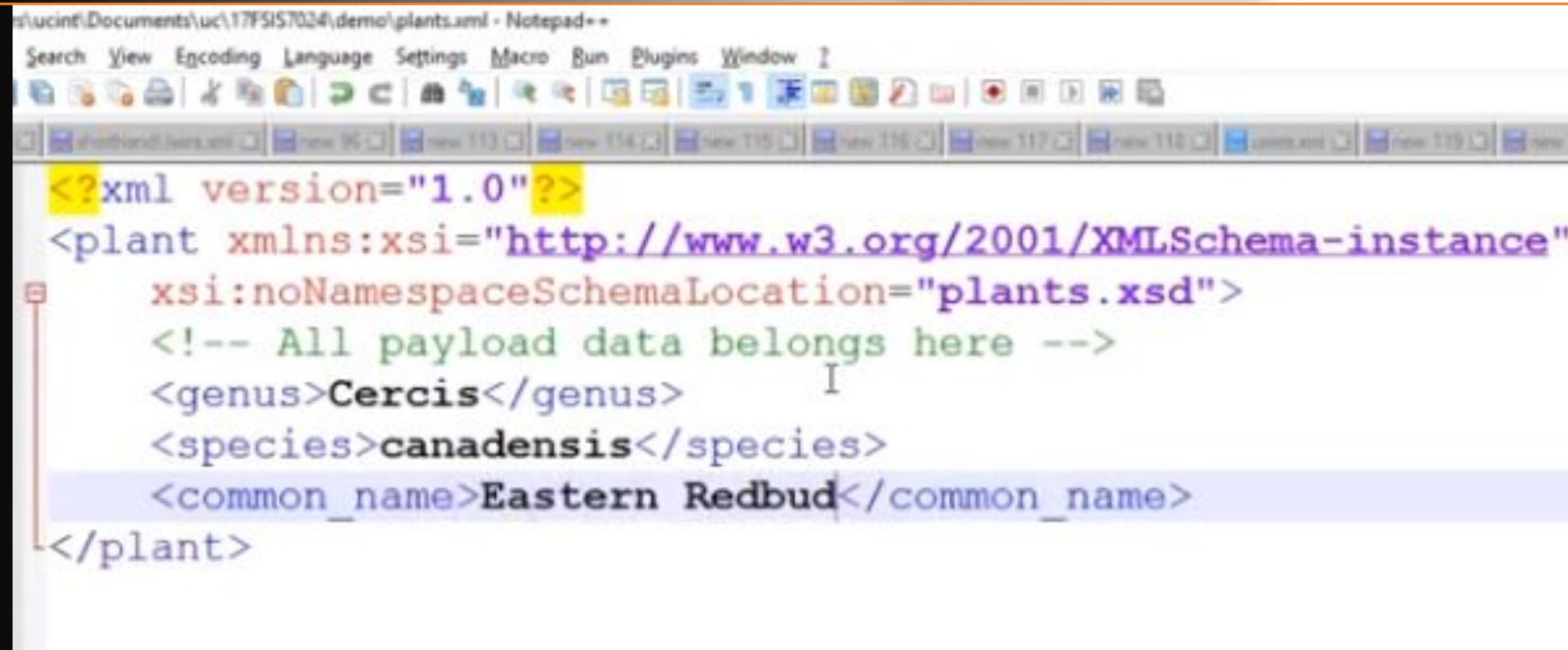
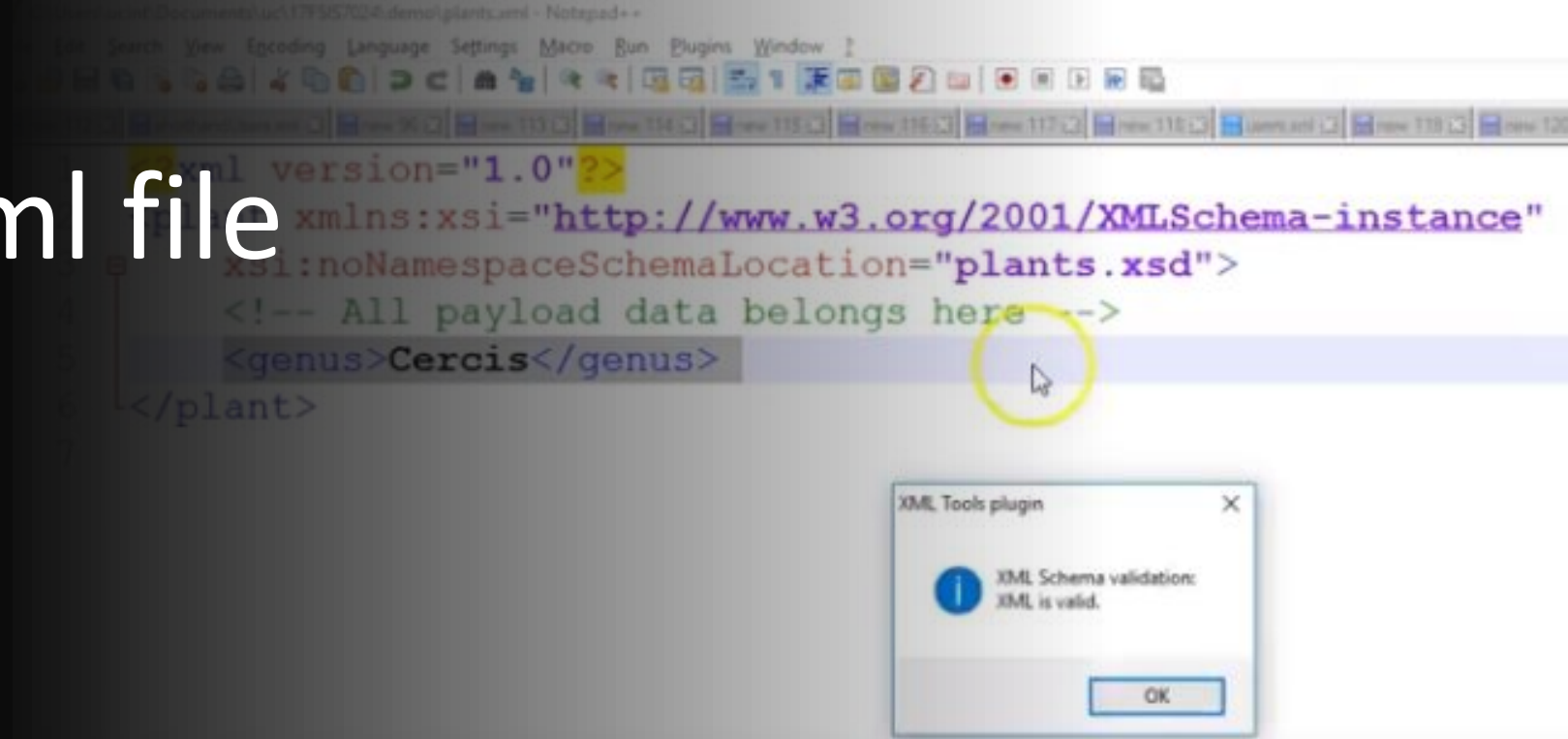
```
1 <?xml version="1.0"?>
2 <plant xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:noNamespaceSchemaLocation="plants.xsd">
4     <!-- All payload data belongs here -->
5     <genus>Cercis</genus>
6 </plant>
7
```

Add more data elements from document definition in xsd file



```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:element name="plant">
4     <xs:complexType>
5       <xs:sequence>
6         <xs:element name="genus" type="xs:string"/>
7         <xs:element name="species" type="xs:string" minOccurs="0"/>
8         <xs:element name="common_name" type="xs:string" minOccurs="0"/>
9         <xs:element name="cultivar" type="xs:string" minOccurs="0"/>
10      </xs:sequence>
11    </xs:complexType>
12  </xs:element>
13 </xs:schema>
```

Add data in xml file
according to
document
definition



PlantPlaces.com accepts plant data in the following forms.



Element/Attribute Name	Description	Required?	Default Value	Type/Restrictions
plant	Root element	1		
_genus	Genus	1..*		Max Length 60
_species	Species	no	spp.	Max Length 60
_cultivar	Cultivated variety (Fuji apple, Golden Delicious apple, etc.)	no		Max Length 60
_height	height, in meters	no	0	integer, max: 100
_begin_bloom_date	date at which the plant begins blooming.	no		
_end_bloom_date	date at which blooms stop	no		
_edible	Does the plant contain edible parts?	no		Boolean
_sun_tolerance	How does the plant tolerate sun?	no		Enumeration: full_sun, part_sun, part_shade, full_shade
_bloom_color	color of flower	no		Enumeration: red, orange, yellow, blue, violet, white
_specimens	Contains a repeating group of specimens.	no		
__specimen	represents an individual specimen	1..*		

Different tag elements for XSD file

- Restriction:

- `<xs:restriction base="xs:string">`
- `<xs:restriction base="xs:integer">`

- Max length=60

`<xs:maxLength value="60"/>`

- Default values

- `<xs:element name="height" type="xs:string" minOccurs="0" default="0"/>`
- `<xs:element name="species" type="xs:string" minOccurs="0" default="spp."/>`

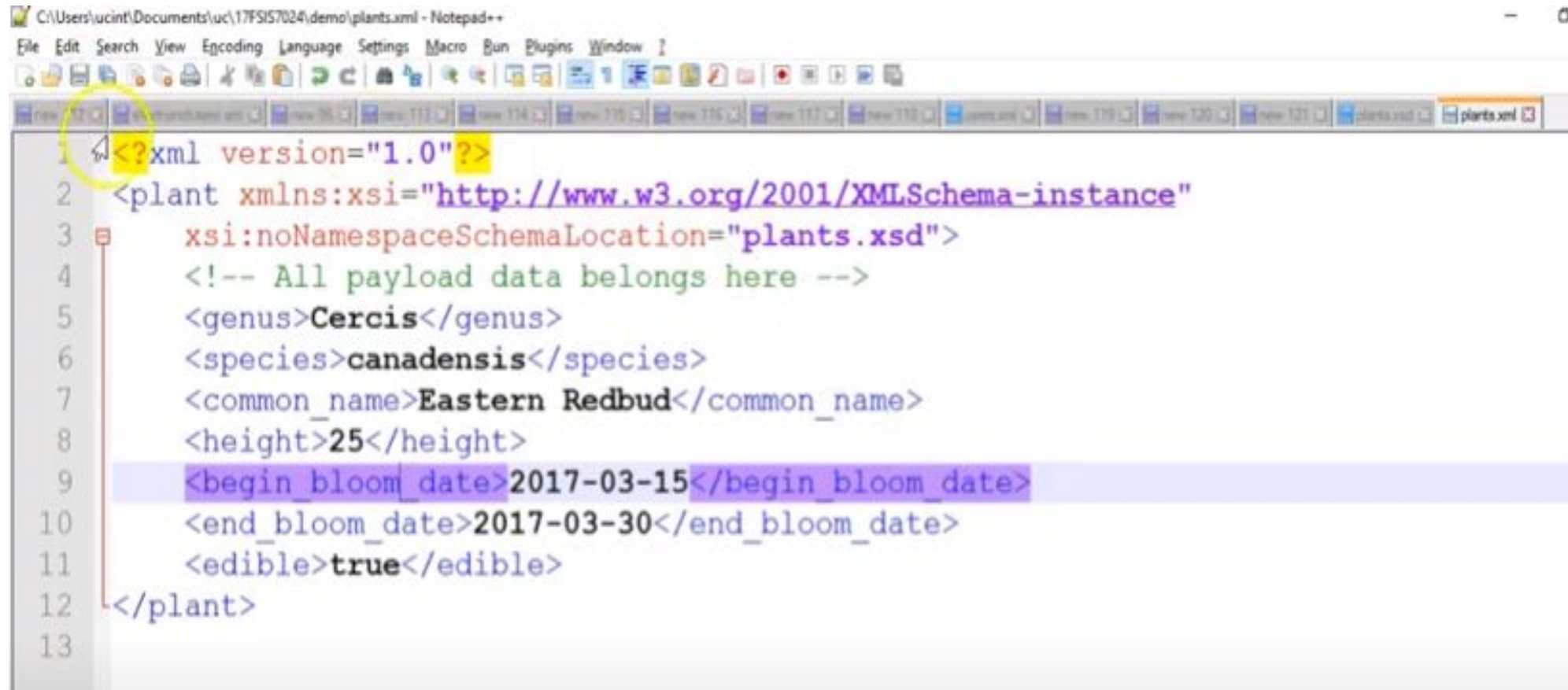
- Date

- `<xs:element name="begin_bloom_date" type="xs:date" minOccurs="0"/>`

- Boolean

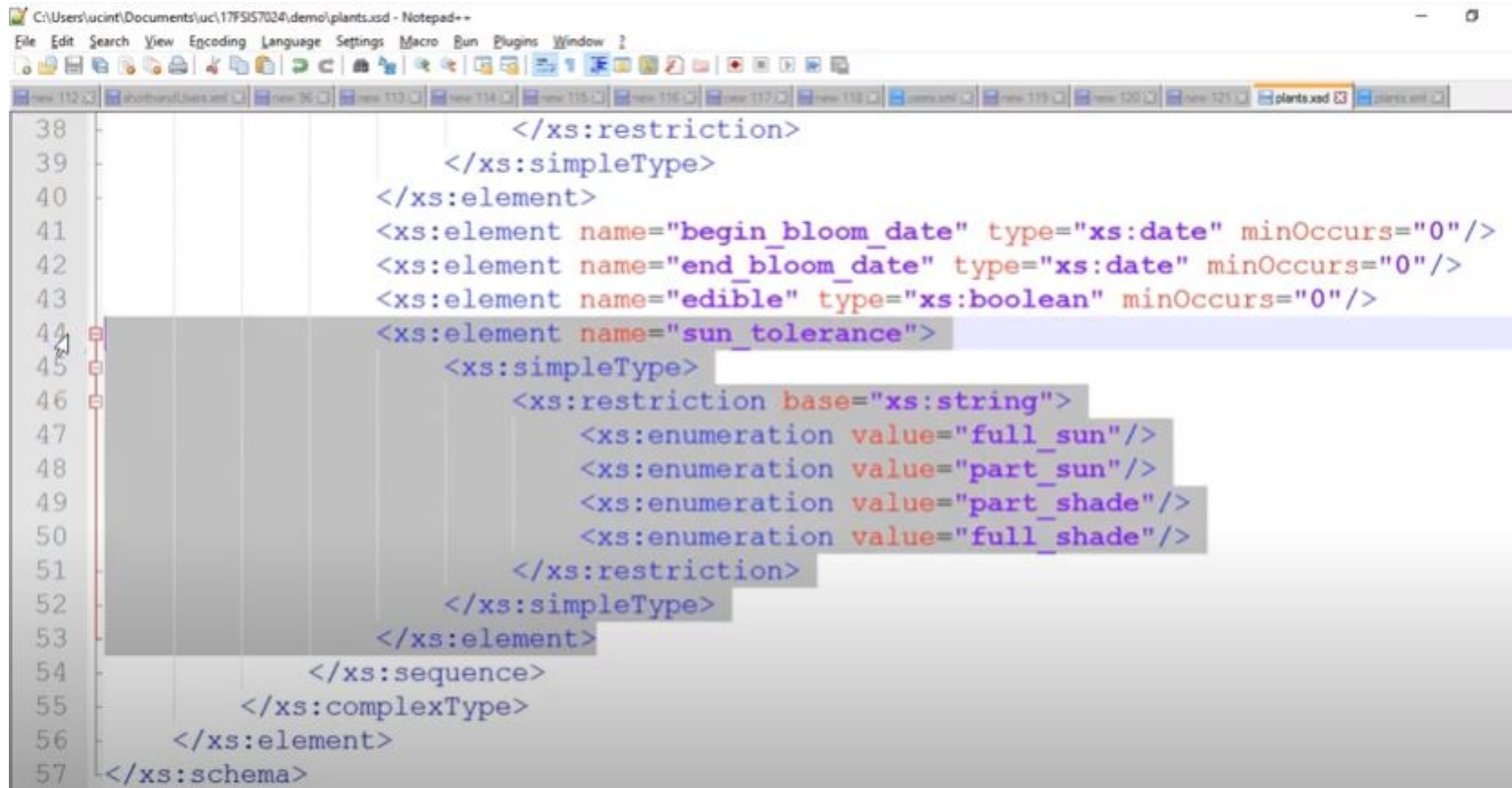
- `<xs:element name="edible" type="xs:boolean" minOccurs="0"/>`

Maintain XML file according to tags defined in XSD file



```
C:\Users\ucint\Documents\uc\17FSIS7024\demo\plants.xml - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window
1 <?xml version="1.0"?>
2 <plant xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:noNamespaceSchemaLocation="plants.xsd">
4   <!-- All payload data belongs here -->
5   <genus>Cercis</genus>
6   <species>canadensis</species>
7   <common_name>Eastern Redbud</common_name>
8   <height>25</height>
9   <begin_bloom_date>2017-03-15</begin_bloom_date>
10  <end_bloom_date>2017-03-30</end_bloom_date>
11  <edible>true</edible>
12 </plant>
13
```

Enumerations define in XSD file



```
C:\Users\ucint\Documents\uc\17FSIS7024\demo\plants.xsd - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window ?
new 112 shortcutUsers.xml new 36 new 113 new 114 new 115 new 116 new 117 new 118 users.xml new 119 new 120 new 121 plants.xsd plants.xml
38         </xs:restriction>
39     </xs:simpleType>
40 </xs:element>
41 <xs:element name="begin_bloom_date" type="xs:date" minOccurs="0"/>
42 <xs:element name="end_bloom_date" type="xs:date" minOccurs="0"/>
43 <xs:element name="edible" type="xs:boolean" minOccurs="0"/>
44 <xs:element name="sun_tolerance">
45     <xs:simpleType>
46         <xs:restriction base="xs:string">
47             <xs:enumeration value="full_sun"/>
48             <xs:enumeration value="part_sun"/>
49             <xs:enumeration value="part_shade"/>
50             <xs:enumeration value="full_shade"/>
51         </xs:restriction>
52     </xs:simpleType>
53 </xs:element>
54 </xs:sequence>
55 </xs:complexType>
56 </xs:element>
57 </xs:schema>
```

Enumeration on “bloom_color” in xsd file

```
C:\Users\ucinf\Documents\uc\17FSIS7024\demo\plants.xsd - Notepad++
File Edit Search View Encoding Language Settings Macro Run Plugins Window
new 112 xshdtransUsers.xml new 36 new 113 new 114 new 115 new 116 new 117 new 118 new 119 new 120 new 121 plants.xsd plants.xml
50         <xs:enumeration value="full_shade"/>
51     </xs:restriction>
52 </xs:simpleType>
53 </xs:element>
54 <xs:element name="bloom_color">
55     <xs:simpleType>
56         <xs:restriction base="xs:string">
57             <xs:enumeration value="red"/>
58             <xs:enumeration value="orange"/>
59             <xs:enumeration value="yellow"/>
60             <xs:enumeration value="blue"/>
61             <xs:enumeration value="violet"/>
62             <xs:enumeration value="white"/>
63         </xs:restriction>
64     </xs:simpleType>
65 </xs:element>
66 </xs:sequence>
67 </xs:complexType>
68 </xs:element>
69 </xs:schema>
```

Add elements of Enumeration in XML file

```
1 <?xml version="1.0"?>
2 <plant xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:noNamespaceSchemaLocation="plants.xsd">
4     <!-- All payload data belongs here -->
5     <genus>Cercis</genus>
6     <species>canadensis</species>
7     <common_name>Eastern Redbud</common_name>
8     <height>25</height>
9     <begin_bloom_date>2017-03-15</begin_bloom_date>
10    <end_bloom_date>2017-03-30</end_bloom_date>
11    <edible>true</edible>
12    <sun_tolerance>full_sun</sun_tolerance>
13    <bloom_color>blue</bloom_color>
14 </plant>
```



THE END